2.2 FFT-Based Circular Convolution Binding

We bundle complex code topologies into a single $O(1)$ state tensor using Vector Symbolic Architecture (VSA). Let $\mathbf{x}_{\text{neural}}$ represent the continuous semantic embedding of a code chunk, and $\mathbf{r}_{\text{symbolic}}$ represent its discrete AST role (e.g., `ROLE_FUNC`). The unified state tensor $\mathbf{T}$ is computed via circular convolution:

$$\mathbf{T} = \sum_{i} \mathbf{r}_{\text{symbolic},i} \circledast \mathbf{x}_{\text{neural},i} \tag{1}$$

This flat 384-dimensional tensor completely rejects structural inversions and distractors without requiring expensive multi-hop graph database traversals.

3 Empirical Evaluation

We evaluated Kalpana-RIF on a 500-query benchmark against the `client-go` repository to measure exact code chunk retrieval accuracy and context compression.

3.1 Context Compression and Precision

Native retrieval pipelines typically pass $K = 5$ chunks to the LLM to maximize recall, incurring heavy token costs. By reranking the $K = 5$ candidate pool using Reciprocal Rank Fusion (RRF) with our discrete state tensors, we successfully compressed the required payload to $K = 3$ while exceeding the original recall ceiling.

Retrieval Configuration	**Top-3 File Accuracy**	**Top-3 Chunk Accuracy**
Miru Native Baseline ($K = 3$)	82.0%	54.8%
Miru + Kalpana Fused ($K = 5 \rightarrow 3$)	**86.3%**	**59.8%**
Absolute Gain	+4.3%	+5.0%

Table 1: 500-query benchmark results demonstrating exact chunk resolution.

3.2 Structural Safety

In production environments, rerankers must not degrade baseline performance. Kalpana-RIF demonstrated near-perfect structural filtering fidelity, achieving 32 wins against 1 single loss ($p = 0.0001$) when compared to the unranked $K = 5$ pool.

4 Conclusion

The Kalpana-RIF Discrete architecture successfully bridges the gap between neural semantic search and discrete AST logic. By achieving a 40% reduction in LLM context requirements with zero latency penalty (< 0.06 ms per pool), it provides a highly scalable, mathematically predictable retrieval foundation for the next generation of agentic coding systems.